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A highly stable and flexible zeolite electrolyte solid-state Li–air battery

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Solid-state lithium (Li)–air batteries are recognized as a next-generation solution for energy storage to address the safety and electrochemical stability issues that are encountered in liquid battery systems[1–4]. However, conventional solid electrolytes are unsuitable for use in solid-state Li–air systems owing to their instability towards lithium metal and/or air, as well as the difficulty in constructing low-resistance interfaces[5]. Here we present an integrated solid-state Li–air battery that contains an ultrathin, high-ion-conductive lithium-ion-exchanged zeolite X (LiX) membrane as the sole solid electrolyte. This electrolyte is integrated with cast lithium as the anode and carbon nanotubes as the cathode using an in situ assembly strategy. Owing to the intrinsic chemical stability of the zeolite, degeneration of the electrolyte from the effects of lithium or air is effectively suppressed. The battery has a capacity of 12,020 milliamp hours per gram of carbon nanotubes, and has a cycle life of 149 cycles at a current density of 500 milliamps per gram and at a capacity of 1,000 milliamp hours per gram. This cycle life is greater than those of batteries based on lithium aluminium germanium phosphate (12 cycles) and organic electrolytes (102 cycles) under the same conditions. The electrochemical performance, flexibility and stability of zeolite-based Li–air batteries confer practical applicability that could extend to other energy-storage systems, such as Li–ion, Na–air and Na–ion batteries.

Li–air batteries have the highest theoretical energy density among existing battery systems and are expected to be prominent in the next generation of energy-storage devices[1,2]. However, several serious challenges concerning safety issues, decomposition and volatilization of the electrolyte, corrosion of the lithium anode and formation of lithium dendrites[3]—which stem from the use of organic electrolytes in conventional Li–air batteries[4–7]—still need to be addressed (Fig. 1a). To circumvent these challenges, it is essential to develop a solid-state Li–air battery (SSLAB), containing a solid electrolyte as the key component[8–10]. In addition to high ionic conductivity, superior stability towards the lithium metal anode and favourable interfacial compatibility, suitable solid electrolytes for SSLABs should have high stability towards the components of air—so that the battery can operate in ambient air[11]—and high resistance to oxidation to prevent corrosion by oxygen-reduction intermediates[12]. However, typical inorganic solid electrolytes that have high ionic conductivity and good safety profiles[13]—including garnets[5], perovskites[11], NASICONs[14] and sulfides[15]—are currently not suitable owing to their instability towards lithium metal and/or air. Furthermore, the high electronic conductivities of solid electrolytes result in the nucleation of lithium inside the electrolytes, which generates short circuits in the batteries[16,17]. In addition, the large-scale production of SSLABs at low cost remains difficult, and the non-flexible nature of conventional inorganic solid electrolytes also restricts their application[18]. To date, the design of a solid electrolyte for SSLABs that simultaneously achieves high environmental adaptability and superior electrochemical performance remains a challenge (Fig. 1a).

Here we report a highly stable, integrated, flexible SSLAB with a lithium-ion-exchanged zeolite X (LiX) zeolite membrane (LiXZM) as the inorganic solid electrolyte (Fig. 1b). LiXZM possesses high ionic conductivity, low electronic conductivity and excellent stability towards the components of air and the lithium anode. In addition, the integrated structure of the SSLAB—which contains carbon nanotubes (CNTs) as the cathode[19]—is favourable for reducing interfacial resistance. The integrated SSLAB exhibits superior performance in ambient air compared with conventional Li–air batteries that contain organic electrolytes. Owing to their desirable structure and excellent performance, Li–air batteries containing solid zeolite electrolytes are expected to find numerous practical applications in energy-storage fields.

Design and characterization of the electrolyte

Zeolites are an important class of inorganic crystalline microporous material that is widely used in the chemical industry[20,21], and those that show high stability towards the components of air meet the fundamental requirement for solid electrolytes in SSLABs. Early studies[22] showed that cations at SII sites (in front of the six-ring windows, inside supercages) and SIII sites (near the four-ring windows, also inside supercages) in zeolite X are the primary contributors to the conduction of ionic carriers. Among the various zeolites, the lithium-ion-exchanged

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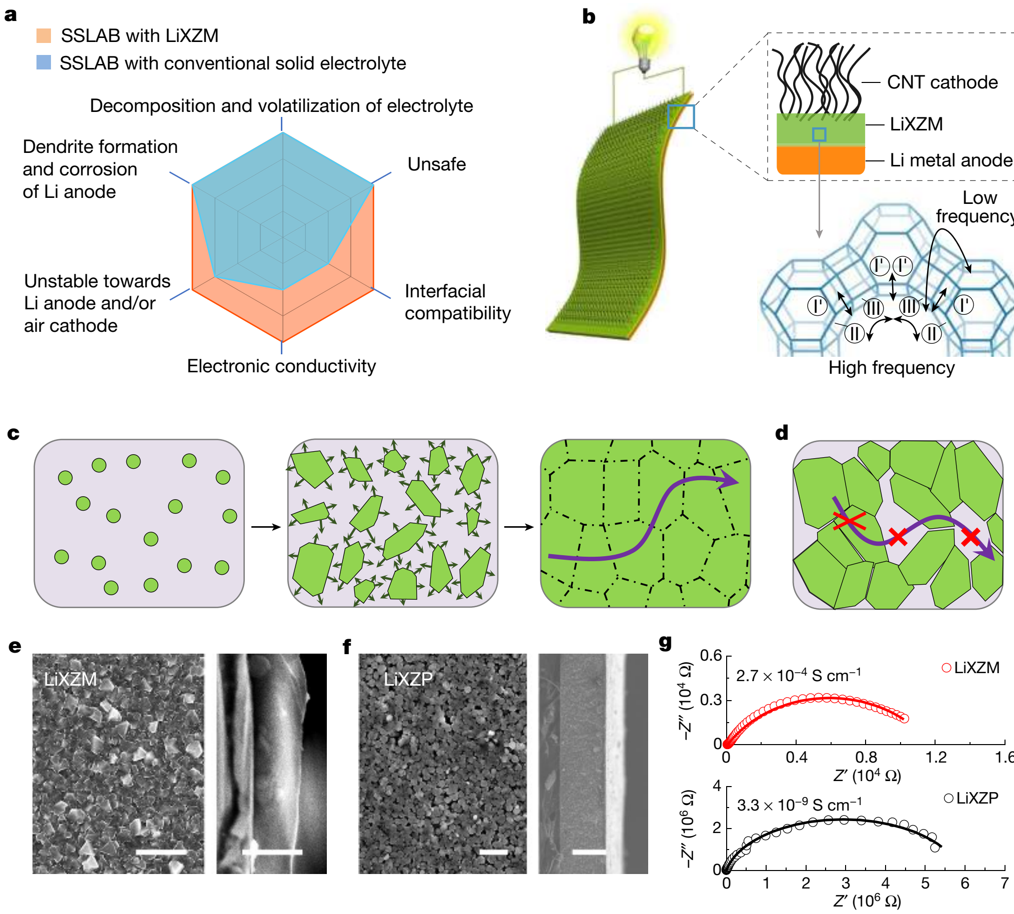

Fig. 1 | Schematic and characterization of the zeolite solid electrolyte. **a**, Radar chart of the capabilities of electrolytes to solve the main problems associated with Li–air batteries. As shown in blue, SSLABs with conventional solid electrolytes can effectively solve several core issues of traditional Li–air batteries with liquid electrolytes. However, a SSLAB with LiXZM as the inorganic solid electrolyte possesses improved stability and interfacial compatibility, and low electronic conductivity, as shown in orange. **b**, Schematic of the integrated SSLAB with C-LiXZM and the conduction mechanism of Li ions in LiX. **c**, **d**, Comparison of the in situ zeolite membrane growth method adopted in this work (**c**) and the traditional preparation method for inorganic solid electrolytes (**d**). **e**, Left, top-view SEM image of LiXZM grown in situ on the stainless-steel sheet using the seed-assisted method. Right, cross-sectional SEM image showing an LiXZM thickness of approximately 5 μm. Scale bars, 10 μm (left); 5 μm (right). **f**, Left, top-view SEM image of LiXZP. Right, cross-sectional SEM image showing an LiXZP thickness of approximately 150 μm. Scale bars, 10 μm (left); 0.1 mm (right). **g**, EIS spectra of LiXZM (top) and LiXZP (bottom). *Z*, impedance.

zeolite X with a low silicon/aluminium (Si/Al) ratio of 1.0 (LiX) has the highest lithium ion content at SIII sites and therefore the highest ionic conductivity[23]. However, LiX has not yet been applied as a solid electrolyte in energy-storage systems owing to the low strength and ionic conductivity of the fabricated zeolite pellets[24–26].

To investigate the intrinsic transport behaviour of lithium ions in LiX, the influence of grain-boundary resistance should be excluded. A lithium-ion-exchanged single crystal[27] with an Si/Al ratio approaching 1.0 (SC-LiX) was prepared to characterize the intrinsic ionic conductivity of this zeolite (Extended Data Fig. 1a). The elemental composition of the sample was analysed by X-ray fluorescence and inductively coupled plasma optical emission spectroscopy (ICP-OES) measurements (Si/Al = 1.03, $Li^+/(Li^+ + Na^+) = 0.98$), and N_2 adsorption–desorption analyses reveal a micropore area of 819 $m^2\,g^{-1}$ and a micropore volume of 0.30 $cm^3\,g^{-1}$ (Extended Data Fig. 1b). After installing silver blocking electrodes on the single crystal under an optical microscope (Extended Data Fig. 1c), the ionic conductivity of SC-LiX was measured in the frequency range of 1 MHz to 10 Hz. Analysis by electrochemical impedance spectroscopy (EIS) (Extended Data Fig. 1d) reveals that the spectrum of SC-LiX features a compressed semicircle, which results from the motion of charge carriers via hopping diffusion along the pores of the zeolite. As shown in Fig. 1b, zeolite X possesses a faujasite (FAU)-type structure, in which cuboctahedral β-cages are connected by double six rings; the cavity structure therefore consists of supercages (α-cages) with diameters of 1.3 nm connected by 12-ring windows with diameters of 0.74 nm. The high-frequency region of the EIS spectrum corresponds to the local transport of lithium ions within α-cages; the cations initially located at the equilibrium SII site move to another SII site across the square elements of SIII by thermally activated hopping. The low-frequency region reflects the long-range-ordered diffusion of cations in the crystal through the windows between adjacent α-cages[28]. Owing to the continuous ion-conduction pathway within a single crystal of zeolite, the conductivity of SC-LiX reaches $2.4 \times 10^{-2}\,S\,cm^{-1}$ (Extended Data Fig. 1d), which is comparable to that of liquid electrolytes[29].

To obtain a homogeneous and dense zeolite-based solid electrolyte for Li–air batteries, the dipcoating and wiping seeded growth[30] methods (Extended Data Fig. 2a) were adopted to prepare an LiXZM on a stainless-steel substrate, as shown in Fig. 1c, e (for X-ray diffraction (XRD) measurements, see Extended Data Fig. 2b). The tightly packed structure ensures smooth ionic migration among the crystals. The absence of a diffraction peak corresponding to the substrate in the XRD pattern of LiXZM indicates its complete coverage on the substrate (Extended Data Fig. 2b), although the thickness of LiXZM is only around 5 μm (Fig. 1e). The crystalline structure of the material is well maintained

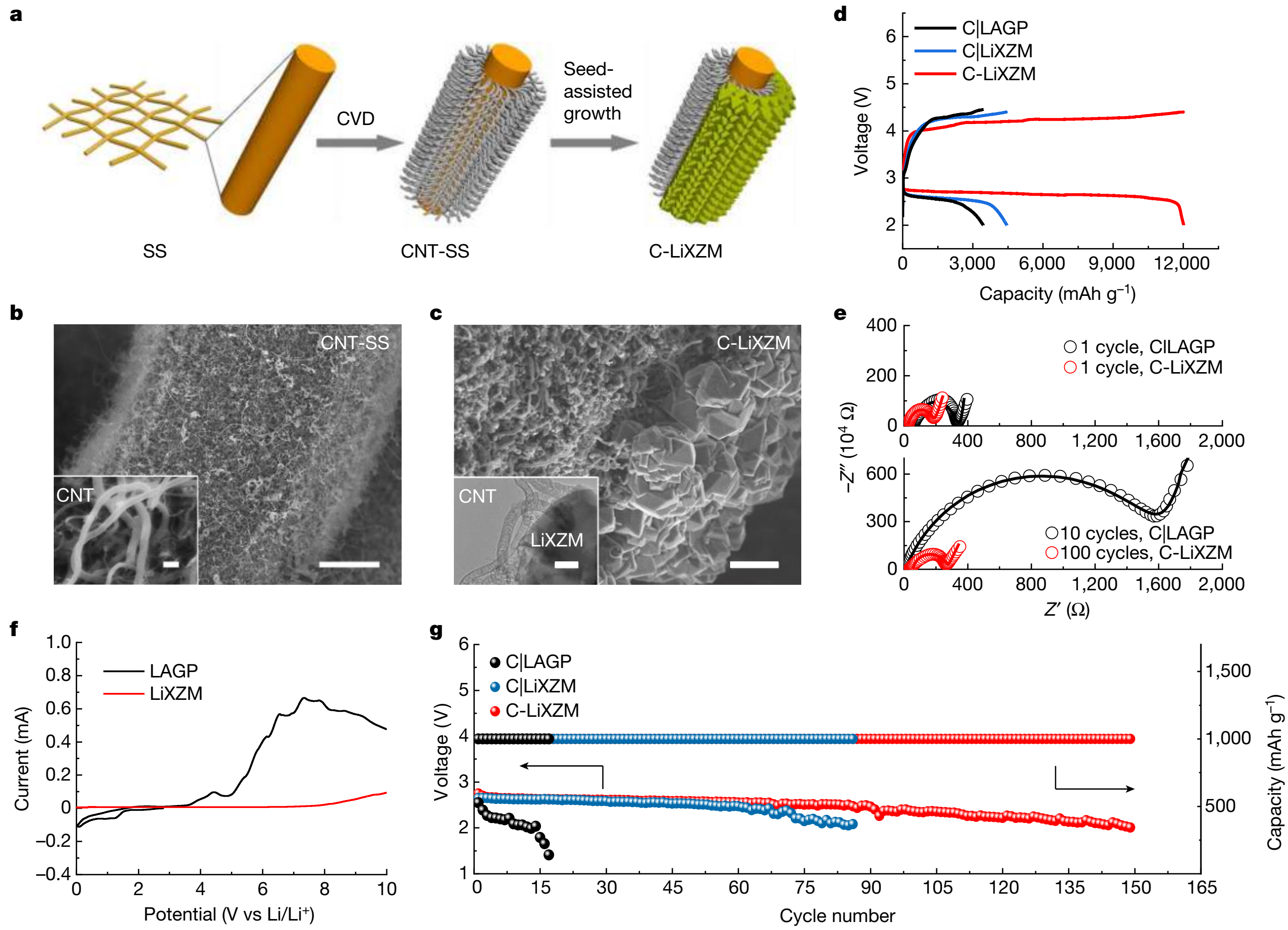

Fig. 2 | Construction and reversibility of the integrated battery with C-LiXZM. a, Schematic of the design and preparation of the integrated C-LiXZM. CNTs were grown in situ on stainless-steel mesh using a chemical vapour deposition (CVD) method under a nitrogen atmosphere. After plasma treatment on one side, the zeolite membrane was grown in situ on the hydrophilic side of CNT-SS using a seed-assisted method. **b**, SEM image of the CNT-SS cathode, with an enlarged image in the inset. Scale bars, 10 µm (main image); 200 nm (inset). **c**, SEM image of the C-LiXZM, with a TEM image of the interface between the CNTs and LiXZM shown in the inset. Scale bars, 2 µm (main image); 100 nm (inset). **d**, Specific capacities of SSLABs with integrated C-LiXZM, non-integrated C|LiXZM and C|LAGP at a current density of 500 mA g^{-1}. **e**, EIS spectra of SSLABs with C-LiXZM and C|LAGP after different numbers of cycles. **f**, Electrochemical windows of LiXZM and LAGP. **g**, Voltage on the discharge terminal of SSLABs plotted against cycle number, at a current density of 500 mA g^{-1} and with a specific capacity limited to 1,000 mAh g^{-1}.

after the ion-exchange process in aqueous solution (Extended Data Fig. 2b), demonstrating the high stability of LiXZM towards water. Notably, no change was observed in the XRD pattern of LiXZM after it was kept under ambient conditions for one year (Extended Data Fig. 2b), which indicates that it is highly stable compared with conventional inorganic solid electrolytes. Such superior stability is crucial for Li–air batteries that are to operate in ambient air. For comparison, an LiX zeolite pellet (LiXZP) was prepared by pressing tablets, a traditional preparation method for inorganic solid electrolytes. The inevitable interspaces among the crystals and the large thickness (around 150 µm) result in a considerable increase of the potential barriers to ionic migration and of the diffusion distance (Fig. 1d, f, Extended Data Fig. 2c). As expected, LiXZM shows a higher ionic conductivity than LiXZP (2.7×10^{-4} S cm^{-1} compared with 3.3×10^{-9} S cm^{-1}, respectively), as shown in Fig. 1g and Extended Data Fig. 2d. The EIS spectra of LiXZM were recorded between 25 °C and 200 °C (Extended Data Fig. 2e, f), and the conduction activation energy (E_a) was calculated to be 6.54 kJ mol^{-1} according to the Arrhenius equation. Notably, LiXZM exhibits excellent electronic insulation, with a stable electronic conductivity of around 1.5×10^{-10} S cm^{-1} at 25 °C–500 °C (Extended Data Fig. 2g, h), which effectively prevents the nucleation and growth of Li dendrites inside the solid electrolyte. LiXZM was found to have an adhesive strength of 28.81 N (measured using a scratch test[31]) and a Vickers hardness of 390.1HV0.05 (where 0.05 indicates the load in kilogram-force) (Extended Data Fig. 2i), whereas LiXZP was too fragile to undergo these experiments. LiXZM, having the better mechanical durability, could therefore effectively prevent the formation and the subsequent negative effects of Li dendrites.

Construction of the LiXZM–electrode interface

It is suggested that rational construction of the interface between a solid LiXZM electrolyte and the cathode might overcome the excessive contact impedance that is observed in batteries with conventional inorganic solid electrolytes. First, as the cathode, nitrogen-doped CNTs were grown in situ on stainless-steel mesh by chemical vapour deposition[19] (CNT-SS; Extended Data Fig. 3a). One side of the CNT-SS was then subjected to plasma treatment to render the CNTs hydrophilic. An NaX zeolite crystal seed slurry was coated on this side (Extended Data Fig. 3b), followed by hydrothermal treatment to form NaXZM. After lithium-ion exchange, the integrated cathode–solid electrolyte structure was obtained (C-LiXZM; Fig. 2a). It is worth noting that the hydrophilic side of CNT-SS is favourable for the dispersal and fitting of the zeolite crystal seed, whereas the hydrophobic side serves as the

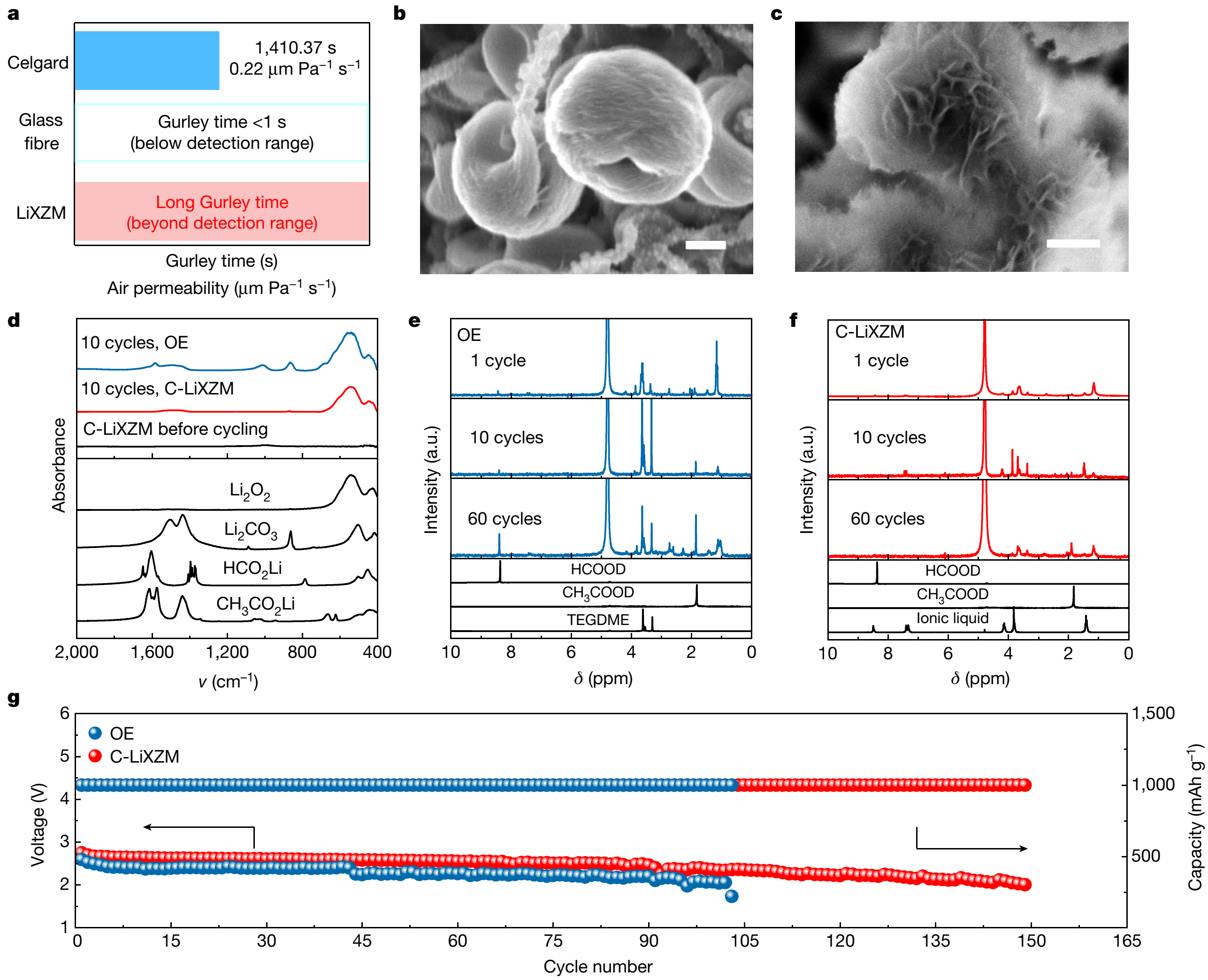

Fig. 3 | Properties of batteries containing C-LiXZM. a, Gurley time and air permeability values for LiXZM, glass fibre and Celgard separators. **b**, **c**, SEM images of the discharge products for the batteries with the organic electrolyte (**b**) and C-LiXZM (**c**) under the same conditions. Scale bar, 200 nm. **d**, FTIR spectra of the discharge products in the batteries with C-LiXZM and the organic electrolyte (OE) after the 10th cycle. **e**, **f**, NMR spectra of the byproducts in batteries with the organic electrolyte (**e**) and C-LiXZM (**f**) after 1, 10 and 60 cycles. **g**, Terminal discharge voltage of batteries with the organic electrolyte and C-LiXZM at a current density of 500 mA g^{-1} with the specific capacity limited to 1,000 mAh g^{-1}.

cathode in contact with air, providing sufficient protection for the discharge products (Li_2O_2) against water erosion. Both surfaces exhibit good affinity for the ionic liquid electrolyte[14,32], which is beneficial both for the further migration of lithium ions towards the cathode and for the protection of Li_2O_2. As shown in Fig. 2b, the CNTs grow evenly on the stainless-steel mesh, and the interwoven spaces created by the CNTs provide adequate space for the transport of oxygen and for the storage of discharge products. The cross-section scanning electron microscopy (SEM) image of the integrated C-LiXZM (Fig. 2c) shows that LiXZM grows in situ on the CNTs at a thickness of about 5 µm (in accordance with Fig. 1e), and that the CNTs retain their shape. The intergrowth of zeolites on the CNTs can be seen in detail in the transmission electron microscopy (TEM) image (Fig. 2c, inset). Raman spectroscopy and X-ray photoelectron spectroscopy (XPS) further demonstrate the structural integrity and the nature of the chemical bonding in C-LiXZM (Extended Data Fig. 3c–f). The thermogravimetric curve of C-LiXZM (Extended Data Fig. 3g) indicates that the mass ratio of CNTs to zeolites is 1:9.45.

Because of the good wettability of lithium metal on LiXZM, the lithium anode was fabricated on the other side of LiXZM via the direct casting of molten lithium[33]. The SEM image shown in Extended Data Fig. 3h confirms the tight and compatible contact between LiXZM and lithium metal. In summary, an integrated cathode–electrolyte–anode system with well-defined interfaces was successfully constructed using a facile battery-assembly process.

Reversibility and stability of the integrated SSLAB

To demonstrate the advantages of the interface constructed between the cathode and the solid electrolyte, non-integrated SSLABs were also assembled using a separate CNT-SS cathode and LiXZM (denoted C|LiXZM; Extended Data Fig. 4a, b). The commercial NASCION-type solid electrolyte $Li_{1.5}Al_{0.5}Ge_{1.5}P_3O_{12}$ (LAGP; Extended Data Fig. 4c) (conductivity 4.5×10^{-4} S cm^{-1}) was also used for comparison. As shown in Fig. 2d, the non-integrated SSLABs based on C|LAGP and C|LiXZM display similar specific capacities of 3,451 and 4,454 mAh g_{CNT}^{-1}, respectively, at a current density of 500 mA g^{-1}, illustrating the potential for LiXZM as a solid electrolyte to replace LAGP in practical applications. However, the reversible capacity of the integrated SSLAB containing C-LiXZM is three times as large (12,020 mAh g_{CNT}^{-1}). The excellent capacity of the integrated battery is a result of the contact between the cathode and the zeolite solid electrolyte, which simultaneously facilitates mass transport and minimizes interfacial resistance. In addition, the

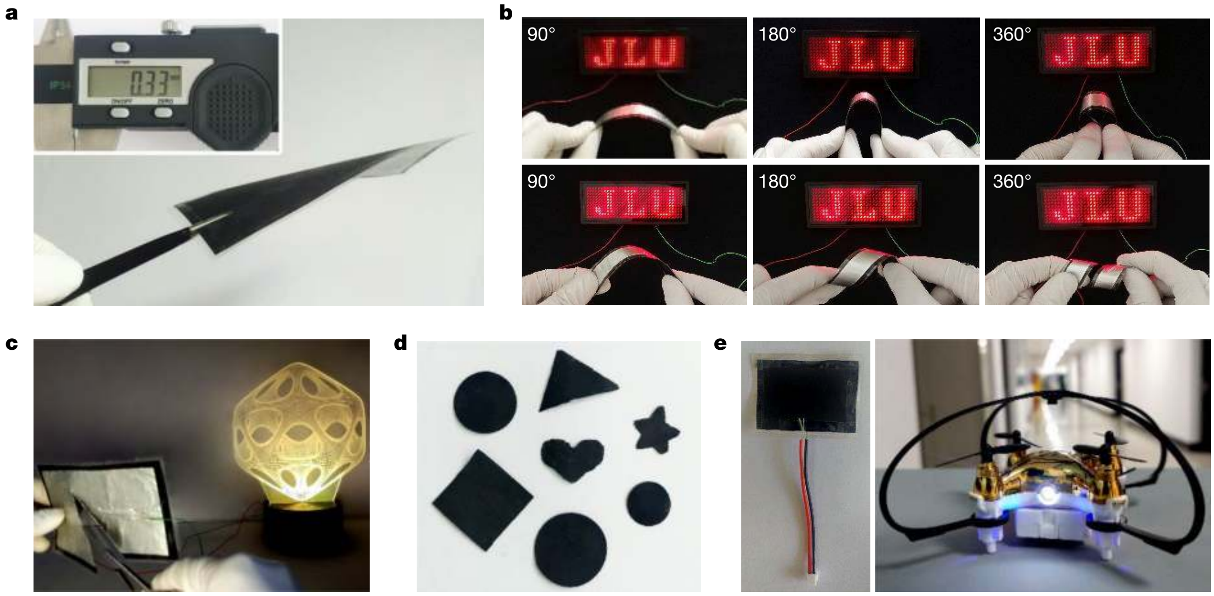

Fig. 4 | Safety, abuse tolerance and flexibility of the SSLAB with C-LiXZM. **a**, The thickness of the SSLAB with C-LiXZM was measured to be 330 μm, and the material displays a bend when picked up with tweezers. **b**, Demonstration of the flexibility of the SSLAB: the thin-strip battery can power a screen of red light-emitting diodes displaying 'JLU' under different bending and torsion conditions. **c**, The working SSLAB with C-LiXZM can continuously operate after being cut and damaged. **d**, The integrated SSLAB with C-LiXZM can be processed into different shapes according to the requirements. **e**, The integrated SSLAB with C-LiXZM powering a small unmanned aerial vehicle.

integrated SSLAB displays a lower overpotential during charging and discharging compared to the batteries based on C|LiXZM and C|LAGP (Extended Data Fig. 4d), demonstrating the superior reversibility of the battery with C-LiXZM. Similar results were obtained at different current densities (Extended Data Fig. 4e), indicating a superior rate performance.

By contrast, the performance of the SSLAB using the LAGP electrolyte gradually declines as the discharge process continues, which is predominantly attributed to the instability of LAGP in the presence of the lithium metal anode. When in contact with the lithium surface, Ge^{4+} in LAGP is readily reduced to Ge^{2+} and Ge^{0}, which leads to degradation of the crystal structure and to decreased lithium-ion conduction and accumulated interfacial impedance during electrochemical processes[14,15] (Extended Data Fig. 5a–l). As a result, the impedance of the SSLAB with the LAGP electrolyte increases markedly after 10 cycles (Fig. 2e), whereas the resistance of the integrated SSLAB with C-LiXZM shows little change even after 100 cycles. Furthermore, LiXZM displays high electrochemical stability in the presence of lithium metal over a wide potential range[34,35] of 0–7 V (Fig. 2f), whereas LAGP is not stable even at the beginning of the electrochemical process (2.8 V). No new product was generated on the surface of LiXZM (Extended Data Fig. 5i–l) or on the lithium metal (Extended Data Fig. 5m, n) after cycling, further confirming the stability of LiXZM in the presence of lithium. In addition, the higher mechanical strength of LiXZM (Extended Data Fig. 5o) compared with LAGP effectively suppresses the growth of lithium dendrites, and thus enhances the cycling performance. The integrated SSLAB containing C-LiXZM can operate for over 149 cycles at a current density of 500 mA g^{-1}, a limited specific capacity of 1,000 mAh g^{-1} and a terminal voltage of 2 V (Fig. 2g). This cycle life is much higher than those of SSLABs with C|LiXZM (86 cycles) and with C|LAGP (12 cycles in this work, with an optimal value of 27 cycles reported for modified LAGP[36]) under the same conditions. The electrochemical performance of SSLABs with C-LiXZM was then measured under various operating conditions and with different mass loadings of CNTs, in order to demonstrate the superiority of the rational design of the SSLAB with C-LiXZM (Extended Data Fig. 6a, b). Furthermore, the much better cyclic stability of SSLAB with C-LiXZM (149 cycles) compared with that of the Li–air battery with only ionic liquid electrolytes (67 cycles) demonstrates that C-LiXZM is predominantly responsible for the enhancement in performance (Extended Data Fig. 6c).

Comparison with conventional battery systems

In general, the electrochemical performances of SSLABs are usually poorer than those of conventional Li–air batteries containing organic electrolytes[19,36]. However, the integrated SSLAB with C-LiXZM displays better a cyclic performance than batteries containing the organic electrolyte (1 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) in tetraethylene glycol dimethyl ether was used in this work) owing to the protection of the anode, the modulation of discharge products and the inhibition of formation of side products. Initially, the resistance to the crossover of air components from the cathode side was evaluated using an air permeability test. Two typical separators in Li–air/O_2 batteries (Celgard and glass fibre) were used in the control groups. As shown in Fig. 3a, the Gurley times for the Celgard (1,410.37 s) and the glass fibre (less than 1 s) separators are not of the same magnitude as that for LiXZM, which was far beyond the detection range. This demonstrates the ultraslow kinetics of air permeation in LiXZM, which prevents anode corrosion[37]. In addition, the time-resolved optical images in Extended Data Fig. 7 show that, once it is protected by LiXZM and C-LiXZM, the lithium metal retains its metallic lustre after 1,000 h of ageing. This illustrates the protective effect of zeolite membranes.

The discharge product of the battery containing an organic electrolyte has a toroidal morphology (Fig. 3b), whereas uniform and slim nanosheet-like discharge products are observed in the SSLAB with C-LiXZM (Fig. 3c). This nanosheet-like morphology—as well as the poor crystallinity (Extended Data Fig. 8a)—ensures the facile decomposition of the discharge products and enhances the reversibility of Li–air batteries[38]. The composition of the discharge product in the SSLAB was determined to be Li_2O_2 using Fourier-transform infrared (FTIR) spectroscopy (Fig. 3d). In addition, the byproducts formed in the batteries containing the organic electrolyte—Li_2CO_3, HCO_2Li and CH_3CO_2Li—are absent in the SSLAB with C-LiXZM, indicating that electrolyte decomposition in traditional Li–air batteries has been mitigated[3]. The negligible differences in the XPS and EIS spectra of LiXZM before and after discharge demonstrate that the pores of the zeolite are not blocked by the discharge products, which could ensure the rapid transfer of lithium ions (Extended Data Fig. 8b, c). In nuclear magnetic resonance (NMR) spectra (Fig. 3e), HCO_2Li and CH_3CO_2Li are detected after initial recharge in the organic electrolyte system, and their levels increase markedly after 60 cycles. By contrast, only a trace amount of CH_3CO_2Li was observed in the LiXZM system after 60 cycles, demonstrating that the SSLAB with C-LiXZM can effectively inhibit side reactions (Fig. 3f). As a result, this system operates steadily for more than 149 cycles at 500 mA g^{-1} and 1,000 mAh g^{-1} (Fig. 3g)—a better performance than that of the Li–air batteries with the organic electrolyte (102 cycles in this work and 120 cycles in the literature[19] under similar conditions). The same conclusion can be reached under other operating conditions (Extended Data Fig. 8d). The failure of the SSLAB with C-LiXZM after 149 cycles is mainly ascribed to the considerable increase in overpotential, caused by cathode passivation as Li_2CO_3 is accumulated during cycling[3] (Extended Data Fig. 8e).

Safety, stability and flexibility of the C-LiXZM SSLAB

A lab-level nail test was performed to evaluate the safety of batteries with C-LiXZM and the organic electrolyte (Extended Data Fig. 9, Supplementary Videos 1, 2). After puncturing, the SSLAB with C-LiXZM operated steadily for more than 100 cycles at 500 mA g^{-1} and 1,000 mAh g^{-1}, whereas the battery with the organic electrolyte was unable to operate (Extended Data Fig. 9b, c). To simulate a real application state with fluctuation, the integrated SSLAB with C-LiXZM was initially operated for 20 cycles (200 mA g^{-1}, 2,000 mAh g^{-1}). After a subsequent interval of 40 days, the battery was operated for 40 cycles under different conditions (500 mA g^{-1}, 1,000 mAh g^{-1}), demonstrating the possibility of using C-LiXZM batteries in practical situations and routine production (Extended Data Fig. 10a).

Owing to their crystalline structures, inorganic solid electrolyte materials are generally fragile, and zeolite crystals are no exception. However, the structural design proposed here for the integrated SSLAB—in which an ultrathin LiXZM with a certain degree of flexibility[39,40] is fully connected to the flexible CNT-SS substrate and the Li anode—results in good flexibility. When picked up using tweezers, the integrated SSLAB with a thickness of about 330 μm has a natural bend (Fig. 4a), indicating the macro-scale flexibility of the battery[41]. Notably, the battery is still able to power light-emitting diodes when twisted by 90°, 180° or 360° (Fig. 4b, Supplementary Video 3), and the ultimate fracture curvature of the battery is 2.55×10^2 m^{-1} (curvature radius 3.92 mm; Extended Data Fig. 10b), which demonstrates its excellent flexibility. The integrated SSLAB continues to function even if it is cut during operation (Fig. 4c). In addition, the tensile strength of C-LiXZM (up to 175 MPa; Extended Data Fig. 10c) is much higher than that of other carbon-based materials, such as carbon cloth and carbon nanofibres[19]. Owing to its excellent shape-tailoring capability, the battery can be customized to various shapes that satisfy different requirements (Fig. 4d), thus broadening its practical applicability. A photograph of large zeolite membranes synthesized in the laboratory is presented in Extended Data Fig. 10d to demonstrate the possibility of scale-up of the fabrication of the integrated C-LiXZM. As shown in Fig. 4e and Supplementary Video 4, the integrated SSLAB with C-LiXZM can provide sufficient power to drive a small unmanned aerial vehicle.

The SSLAB with C-LiXZM can achieve a high energy density of about 662 Wh kg^{-1} based on the weight of the components—including active materials of the cathode and anode, ionic liquid and zeolite electrolytes, without considering the casting materials and the substrate. It is expected that the application of zeolite membranes as solid electrolytes can be extended to other energy-storage devices. Because sodium ions have been identified as cationic species with low ion conduction activation energy in some zeolites[26] and skeleton structures[42], the integrated structure proposed in this work is expected to be a promising alternative for solid-state Na–air batteries and Na–ion batteries. In addition, because the conduction of divalent ions in the zeolite is feasible[43], zeolite-based solid electrolytes could be applied in calcium–ion and magnesium–ion batteries. As for a solid-state thin-film device, bipolar stacking of the anode of one battery and the cathode of the next on the same current collector can be implemented to obtain higher voltage and energy density[11,44,45]. This fabrication strategy is expected to expand the practical application of battery systems that have industrialized cathodes and anodes, such as Li–ion and Na–ion batteries.

In summary, through material selection and structural design, we have developed a highly stable, flexible and integrated solid-state Li–air battery with an LiX zeolite membrane as the inorganic solid electrolyte. This material overcomes the disadvantages of conventional solid electrolytes, including instability, poor interfacial compatibility and high electronic conductivity. The battery shows high capacity and rate capability and a long cycle life in ambient air. Owing to their electrochemical performance, safety, flexibility and environmental adaptability, these systems may hold promise for next-generation energy storage.

Online content

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Methods

Chemicals and materials

All chemicals were directly used without further purification. Fumed silica was obtained from Xuzhou Tiancheng Chlor-alkali Co. Triethanolamine (TEA, ≥99%) and NaOH were purchased from Beijing Chemical Works. Na_2SiO_3·$9H_2O$ and LiCl were obtained from Tianjin Guangfu Chemical Reagent Co. $NaAlO_2$ (Al_2O_3, 41%) was purchased from Sinopharm Chemical Reagent Co. Stainless-steel mesh (304) was purchased from Changzhou Hangshuo Mesh Materials Co. Melamine was purchased from Sinopharm Chemical Reagent Co. $Li_{1.5}Al_{0.5}Ge_{1.5}P_3O_{12}$ (LAGP) was purchased from Shenzhen Kejing Star Technology Co. Li_2CO_3, CH_3CO_2Li and HCO_2Li were purchased from Sigma-Aldrich. Li_2O_2 was purchased from Acros. 1-Ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) ([C_2C_1im][NTf_2]), tetraethylene glycol dimethyl ether (TEGDME), lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) and D_2O were purchased from Aladdin Reagent.

Synthesis of SC-LiX

The synthesis of SC-LiX was based on a modified previously reported method[27], to yield a product with the molar composition of 4.0 Na_2O: 1.0 Al_2O_3: 2.5 SiO_2: 440 H_2O: 7.0 TEA. Specifically, the silica precursor solution was prepared with fumed silica slurry, whereas the sodium aluminate precursor solution was prepared by adding sodium aluminate into sodium hydroxide solution. Subsequently, the sodium aluminate precursor solution was filtered with a 0.45-μm membrane filter, followed by the addition of TEA. After keeping at room temperature for 1 day, the silica and TEA-containing sodium aluminate precursor solutions were mixed in a 100-ml Teflon-lined stainless-steel autoclave, and then crystallized at 95 °C for 20 days. The products were vacuum filtered, and dried at 60 °C for 12 h. Finally, ion exchange was carried out with an appropriate amount of LiCl solution at 50 °C, at a rotation rate of 30 rpm, for a total of 6 times (once in 0.5 M LiCl solution, twice in 1 M solution, twice in 2 M solution, then finally in 3 M solution), and the solution was renewed every 8 h. The products were repeatedly washed with water and ethyl alcohol, and stored in a desiccator after drying at 60 °C for 12 h. Samples were dehydrated at 300 °C for 12 h before characterization and use. Elemental analysis of SC-LiX (ICP-OES): Si 51.54%, Al 48.46%, Si/Al 1.03; Li 94.98%, Na 5.02%, $Li^+/(Li^++Na^+)$ 0.98. Elemental analysis of SC-LiX (X-ray fluorescence (XRF)): Si 24.92%, Al 24.37%, Si/Al 0.99.

Synthesis of NaX zeolite seeds

NaX zeolite seeds were synthesized using a conventional hydrothermal process with the molar composition of 3.58 Na_2O: 1.0 Al_2O_3: 2.24 SiO_2: 171.18 H_2O. First, the mixture was prepared by mixing sodium silicate solution and sodium aluminate solution followed by continuous stirring for 100 h. After transferring into a 50-ml Teflon-lined stainless-steel autoclave, the mixture was placed in the oven at 75 °C for 5 h and then at 95 °C for 2.5 h. Finally, the products were washed by centrifugation, dried at 60 °C for 12 h, and stored in a desiccator.

Synthesis of LiXZM

A dipcoating and wiping seeded growth method was adopted to prepare LiXZM[30]. First, the as-obtained zeolite NaX seeds were dispersed in deionized water to prepare the seed slurry, and the slurry was evenly coated on the substrate by direct but very careful rubbing of the seed slurry onto the substrate. Then, the seed-pretreated substrate was dried at 60 °C for 1 h, followed by heating at 200 °C for 1 h. After cooling to ambient temperature, the substrate was placed in the synthetic gel with a molar composition of 3.58 Na_2O: 1.0 Al_2O_3: 2.24 SiO_2: 171.18 H_2O, and heated at 95 °C for 4 h. Afterwards, the products were washed by ultrasound treatment and dried at 60 °C for 12 h. Finally, ion exchange was carried out using the procedure mentioned above (see section 'Synthesis of SC-LiX' for SC-LiX), except a rotation rate of 50 rpm was used. Samples were dehydrated at 300 °C for 12 h before characterization and use. Elemental analysis of LiXZM (ICP-OES): Si 53.44%, Al 46.56%, Si/Al 1.11; Li 95.37%, Na 4.63%, $Li^+/(Li^++Na^+)$ 0.99. Elemental analysis of LiXZM (XRF) : Si 24.72%, Al 24.61%, Si/Al 0.97.

Preparation of LiXZP

LiXZP was prepared according to a well-established method for inorganic solid electrolytes. After the zeolite X membrane was removed from the Teflon-lined stainless-steel autoclave in the synthetic procedure of LiXZM, the residual zeolite X powder was washed by centrifugation and dried at 60 °C for 12 h. The ion exchange was then carried out using the same procedure as for LiXZM. Finally, the obtained powder was pressed into pellets with a 10-mm diameter and calcined at 400 °C for 12 h. Samples were dehydrated at 300 °C for 12 h before characterization and use.

Preparation of the integrated C-LiXZM

First, nitrogen-doped carbon nanotubes as the cathode were grown in situ on 304 stainless-steel mesh (1,000 meshes) by chemical vapour deposition (CNT-SS)[19] (wettability of water and ionic liquid electrolyte (IL) on the surface of CNT-SS: $\theta_{water} = 149.2 \pm 2.7°$, $\theta_{IL} = 0°$). One side of the CNT-SS was coated with a crystal seed slurry after a 300-s plasma treatment (wettability of water and ionic liquid electrolyte on CNT-SS used for the growth of zeolite: $\theta_{water} = 36.1 \pm 0.7°$, $\theta_{IL} = 0°$), and the zeolite membrane (wettability of water and IL on zeolite membrane: $\theta_{water} = 0°$, $\theta_{IL} = 31.5 \pm 0.4°$) was grown on this side with a seed-assisted method as described above (see section 'Synthesis of LiXZM') while the other side was covered. The ion exchange was then carried out under argon protection using the same procedure as for LiXZM. Samples were dehydrated at 300 °C under an argon atmosphere for 12 h before characterization and use.

Fabrication of the integrated SSLAB with C-LiXZM

The molten lithium (caution!) was cast directly as the anode on the other side of the zeolite membrane in a glove box under Ar atmosphere, and ionic liquid electrolyte (IL, 1 μL cm^{-2}) was evenly and gently smeared with a tiny brush on the cathode side of C-LiXZM. Finally, the battery was placed in vacuum for 30 min, so that the IL would form a uniform liquid thin film on the surface of CNTs. The IL was 0.5 M LiTFSI dissolved in [C_2C_1im][NTf_2]. The energy density of SSLAB with C-LiXZM is up to 662 Wh kg^{-1} based on the weight of the components including active materials of cathode ($m_{CNT} = 0.0002$ g) and anode ($m_{Li} = 0.0042$ g), IL ($m_{IL} = 0.0031$ g), and zeolite electrolyte ($m_{LiXZM} = 0.0021$ g), without considering the casting materials (stainless-steel battery case) and the substrate (stainless-steel mesh) as the usual estimation at the lab level. The value of capacity and voltage are 2.40 mAh and 2.65 V, respectively.

Fabrication of the non-integrated SSLAB with C|LiXZM and LAGP

The molten Li was cast directly as the anode on LiXZM and LAGP, and the IL (1 μl cm^{-2}) was evenly and gently smeared with a very small brush on the separate CNT-SS cathode. Finally, the battery was placed under vacuum for 30 min, so that the IL would form a uniform liquid thin film on the surface of CNTs.

Fabrication of the battery with organic electrolyte

The battery containing organic electrolyte was assembled with a lithium metal anode, a glass fibre separator, and a CNT-SS cathode with 100 μl organic electrolyte containing 1 M LiTFSI in TEGDME.

Characterization techniques

XRD patterns were obtained on a RigakuD/max2550/PCX-ray diffractometer using Cu Kα radiation. SEM and the corresponding energy-dispersive X-ray spectroscopy (EDS) images were obtained on a JEOL JSM-7800F electron microscope, and TEM images were performed on a Tecnai F20 electron microscope. The micrograph was obtained on a Sky policy optics microscope. N_2 adsorption–desorption isotherms were measured on a Micromeritics ASAP-2020. Vicker hardness was measured using Buehler Micromet 5103 Vickers microhardness instrument. The scratch test was performed using a CSM large-load scratch meter. The thermogravimetric curve was obtained on a TGA Q500 instrument from room temperature to 800 °C in air, at a heating rate of 10 °C min^{-1}. The Gurley time is defined as the time required for 100 cm^3 of air to pass through a separator under an air pressure of 0.862 kgf cm^{-2}. ^1H NMR spectra were recorded on a Varian 300 MHz NMR spectrometer. XPS was performed using an ESCALAB 250 spectrometer. Li–air batteries were tested on a LAND CT2001A multi-channel battery testing system. Electrochemical impedance spectroscopy was performed on a CHI660 Electrochemical workstation. The current density and the specific capacity in this paper were calculated on the basis of the mass of CNTs.

Data availability

The data generated in this study are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

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Author contributions J.Y. and J.X. conceived the project. X.C., J.Y., J.X., J.D. and M.L. designed the experiments. X.C. prepared materials, performed measurements and analysed the data. P.B., L.S., X.W., F.L. and S.L. helped with some of the experiments and characterization. All authors discussed the results and commented on the manuscript. X.C. and M.L. wrote the draft, and J.X. and J.Y. revised and finalized the manuscript.

Competing interests The authors declare no competing interests.

Additional information

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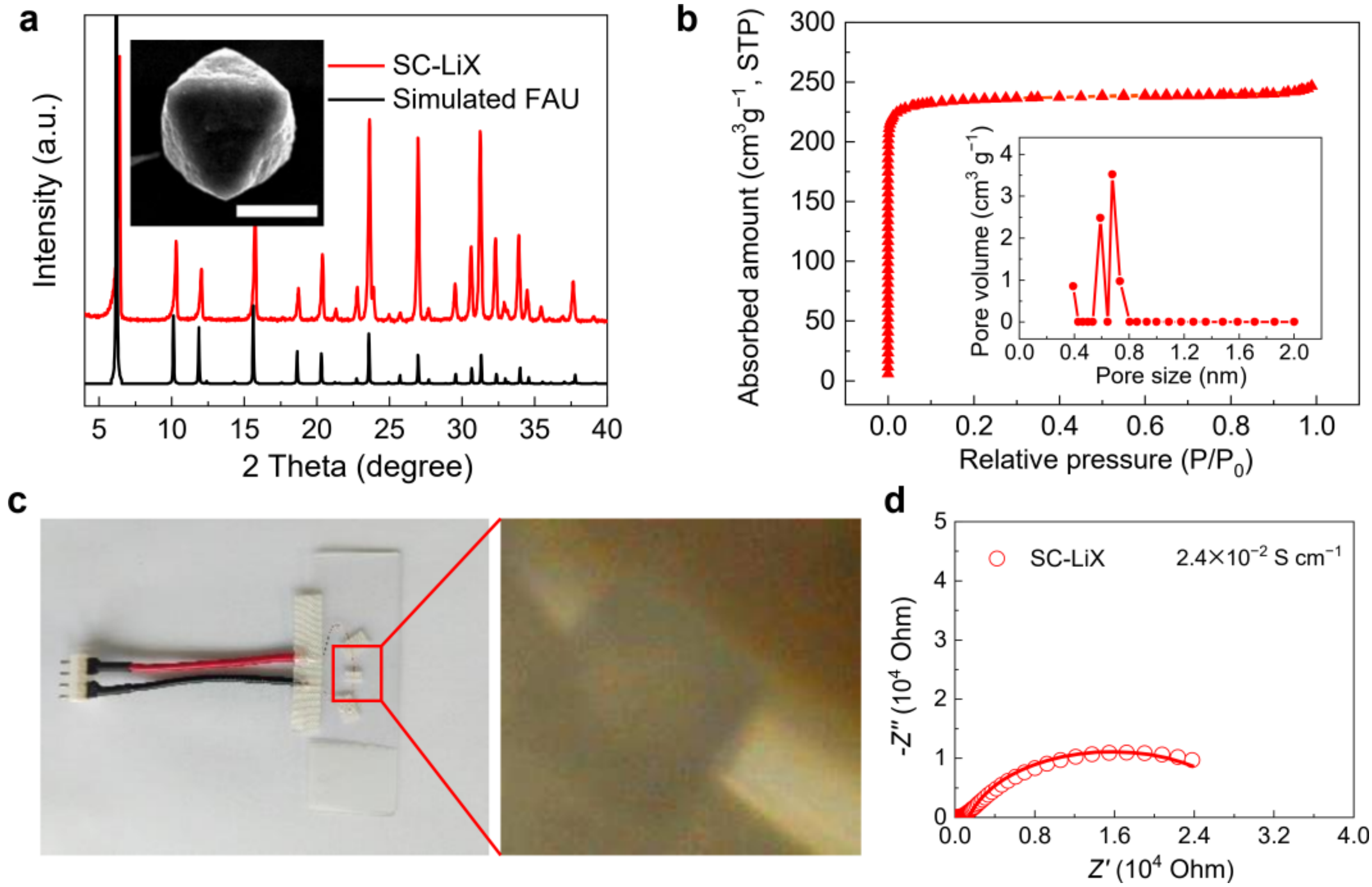
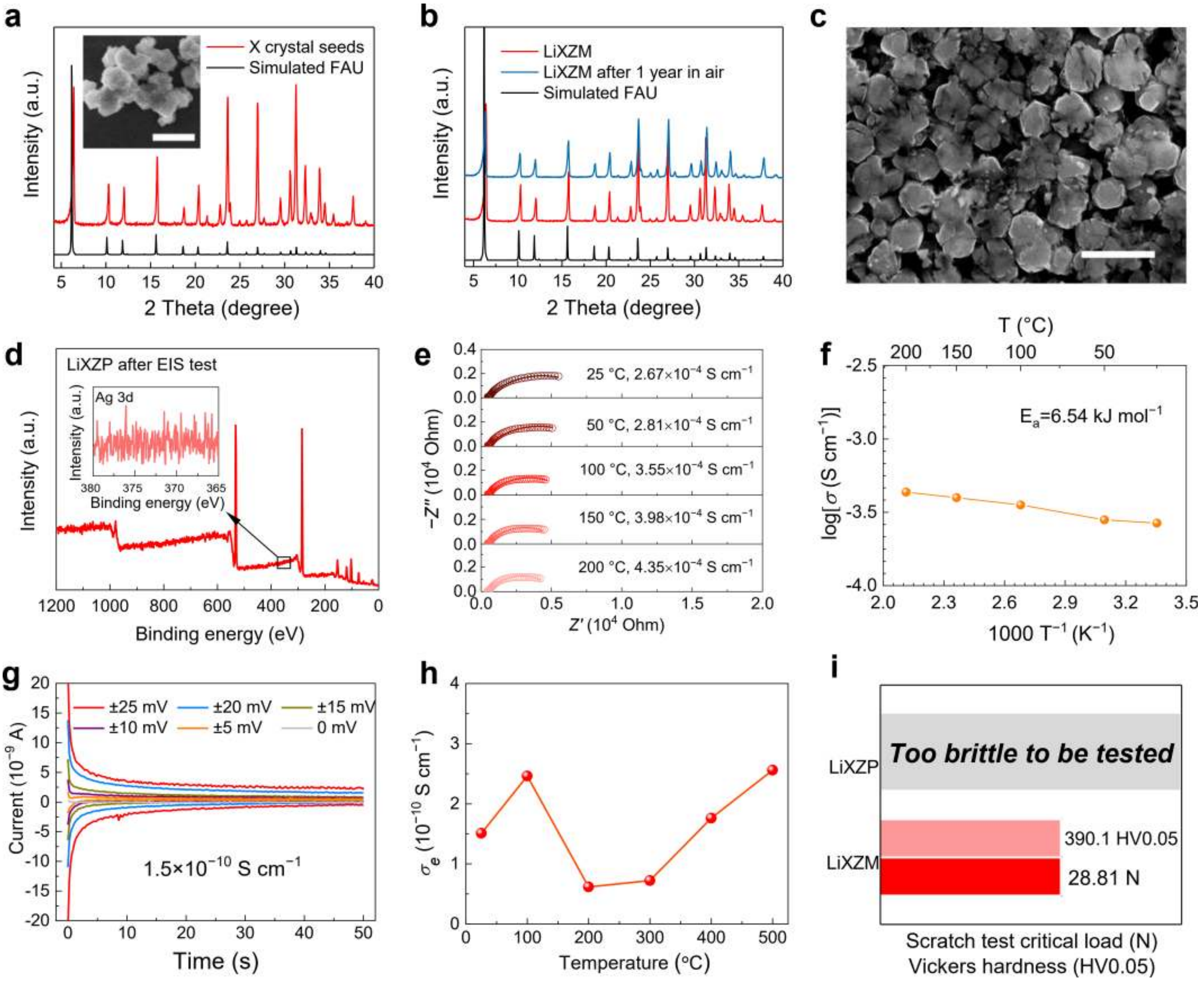
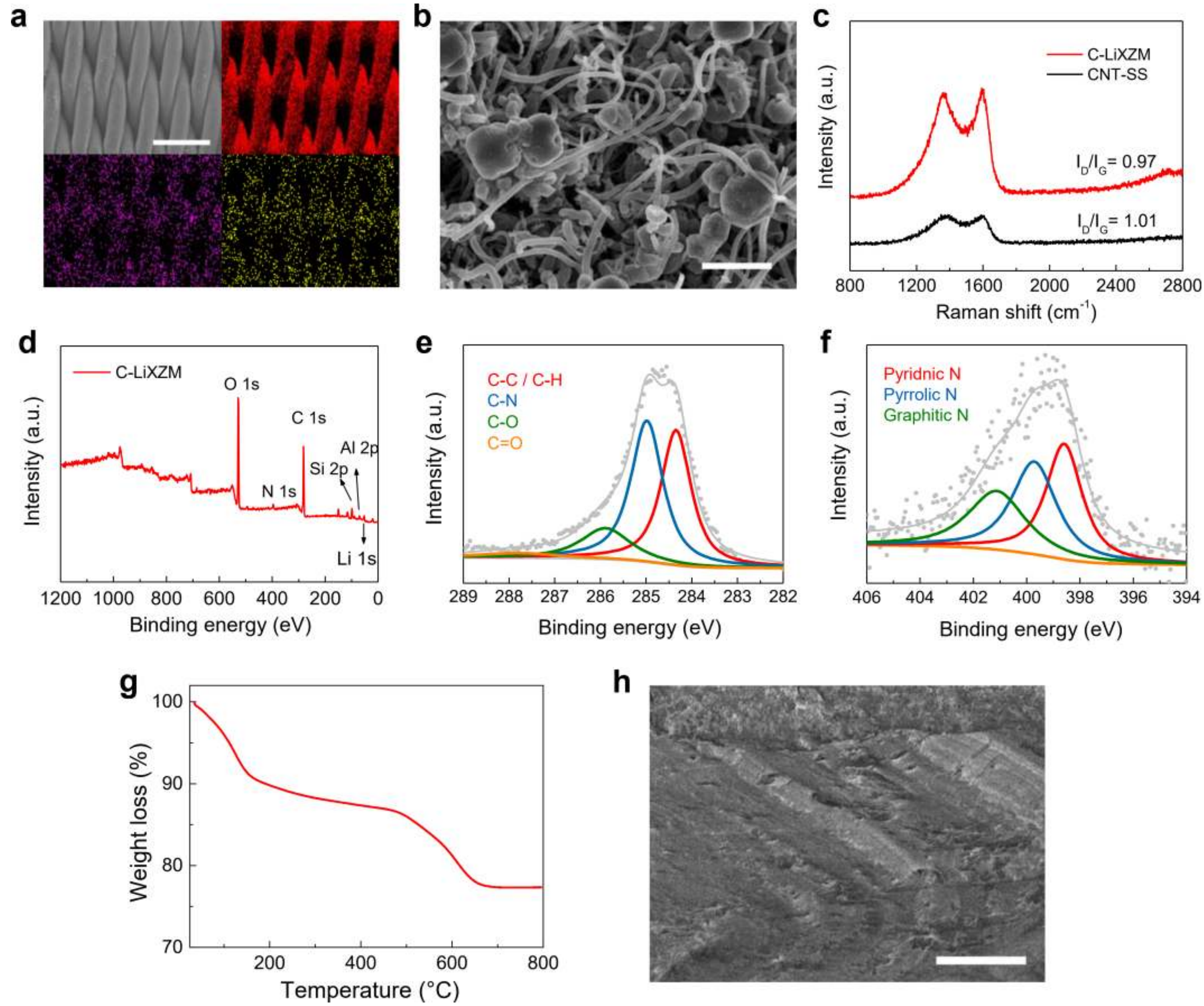


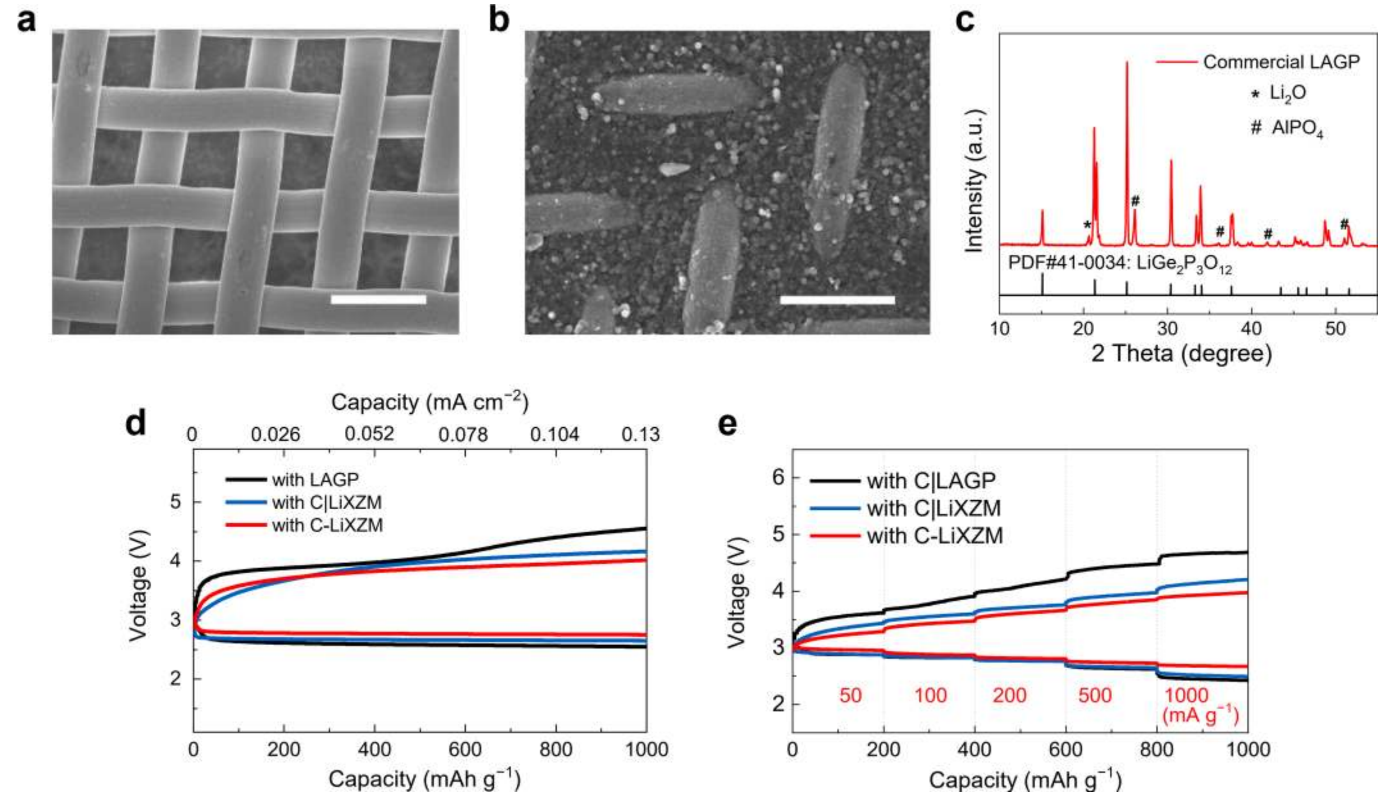
Extended Data Fig. 1 | Characterization of SC-LiX. a, XRD patterns and SEM image (scale bar, 20 µm) of the synthesized SC-LiX. The standard XRD pattern of the simulated FAU structure is provided for comparison. The SC-LiX displays a typical octahedral morphology with a particle size of 28 µm. **b**, N_2 adsorption–desorption isotherms and pore-size distributions of SC-LiX (total surface area = 880 $m^2\,g^{-1}$, micropore area = 819 $m^2\,g^{-1}$, external surface area = 61 $m^2\,g^{-1}$, micropore volume = 0.30 $cm^3\,g^{-1}$). **c**, Photograph and micrograph of the test device used for the EIS of SC-LiX. The ionic conductivity was calculated with the formula $\sigma = d/(R \times S)$, in which the value of the resistance R (28,656 Ω) was obtained from the fitted Nyquist plots, the value of the diameter d (2.3×10^{-3} cm) was measured from the SEM image in the inset of **a**, and S (3.4×10^{-6} cm^2) was the adhesion area of the silver electrode for SC-LiX, with the shape of an equilateral triangle. **d**, EIS spectrum of the synthesized SC-LiX.



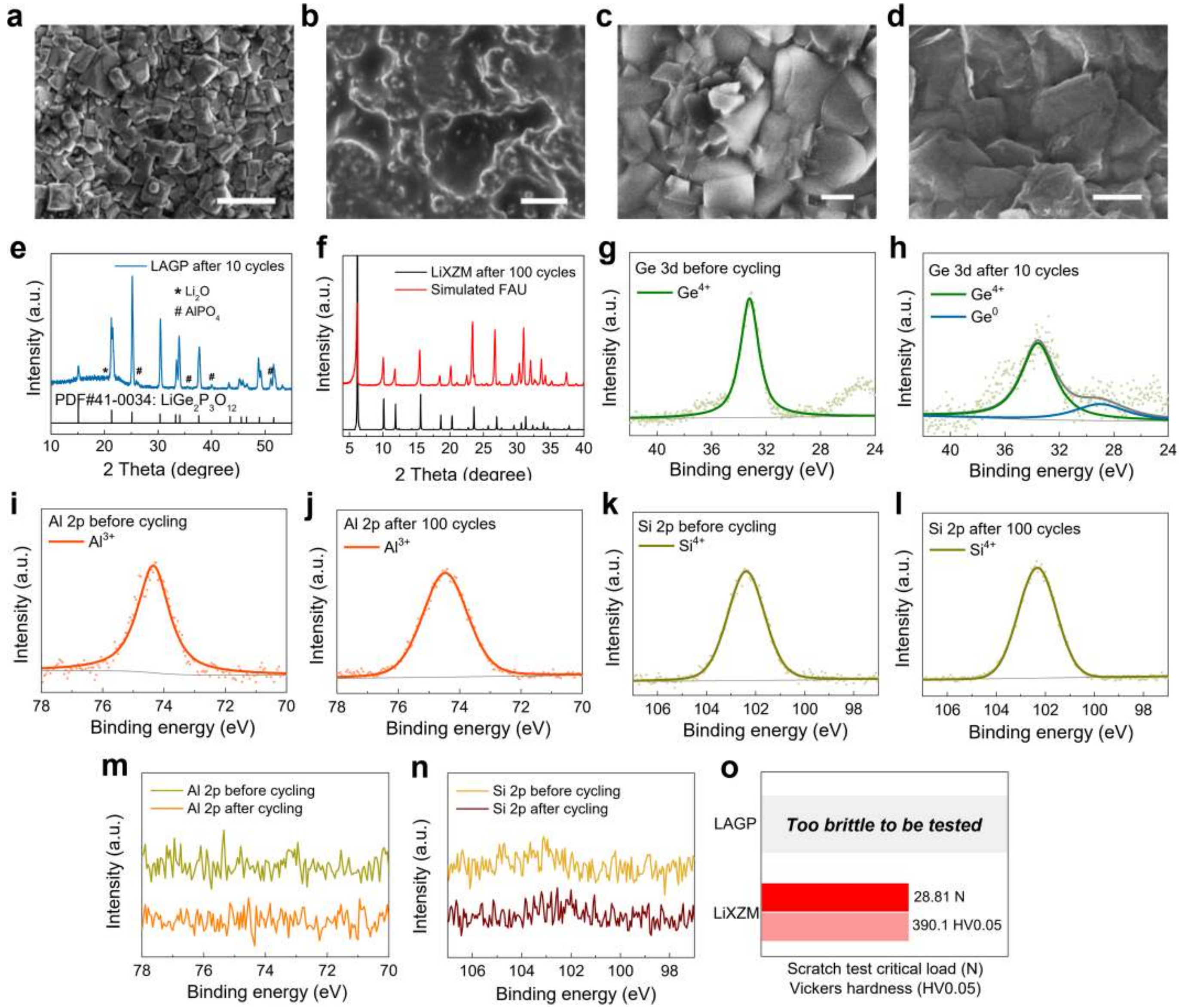
Extended Data Fig. 2 | Characterization of zeolite-based solid electrolyte. **a**, XRD pattern and SEM image (scale bar, 1 μm) of the crystal seeds used in the synthesis of LiXZM. Crystal seeds display a particle size of about 800 nm. **b**, XRD patterns of LiXZM before and after keeping for 1 year. **c**, SEM image of LiXZP (scale bar, 5 μm). **d**, XPS spectrum and Ag 3*d* spectra of LiXZP after the EIS experiment. After EIS, the XPS spectrum of LiXZP was conducted after scraping off the Ag current collectors from the LiXZP. No Ag could be detected in the Ag 3*d* spectrum, demonstrating that Ag could not participate in ion conduction as a mobile ion. The ionic conductivities of LiXZM and LiXZP were calculated using the formula $\sigma = d/(R \times S)$, in which the values of R ($5.9 \times 10^6\ \Omega$ for LiXZP, 11,720 Ω for LiXZM) were obtained from the fitted Nyquist plots, the values of d (1.5×10^{-2} cm for LiXZP, 5.0×10^{-4} cm for LiXZM) were from the SEM images in Fig. 1e, f, and the values of S ($7.9 \times 10^{-1}\ \mathrm{cm}^2$ for LiXZP, $1.6 \times 10^{-4}\ \mathrm{cm}^2$ for LiXZM) corresponded to the adhesion area of silver electrodes for LiXZP (circle) and LiXZM (rectangle, stripped from the smooth stainless-steel substrate). Thus, the ionic conductivities were calculated to be $3.3 \times 10^{-9}\ \mathrm{S\ cm^{-1}}$ for LiXZP and $2.7 \times 10^{-4}\ \mathrm{S\ cm^{-1}}$ for LiXZM. **e**, **f**, EIS spectra (**e**) and the corresponding Arrhenius conductivity plots (**f**) of LiXZM at 25–200 °C in the frequency range of 1 MHz–10 Hz. According to the Arrhenius equation ($\sigma = \sigma_0 \exp(-E_a/k_B T)$)[34], the activation energy of conduction (E_a) is calculated to be 6.54 kJ mol^{-1}. **g**, **h**, Current–time curves (**g**) and the corresponding electronic conductivity (**h**) of LiXZM at 25–500 °C. **i**, Results of scratch and Vickers hardness tests on LiXZM and LiXZP.



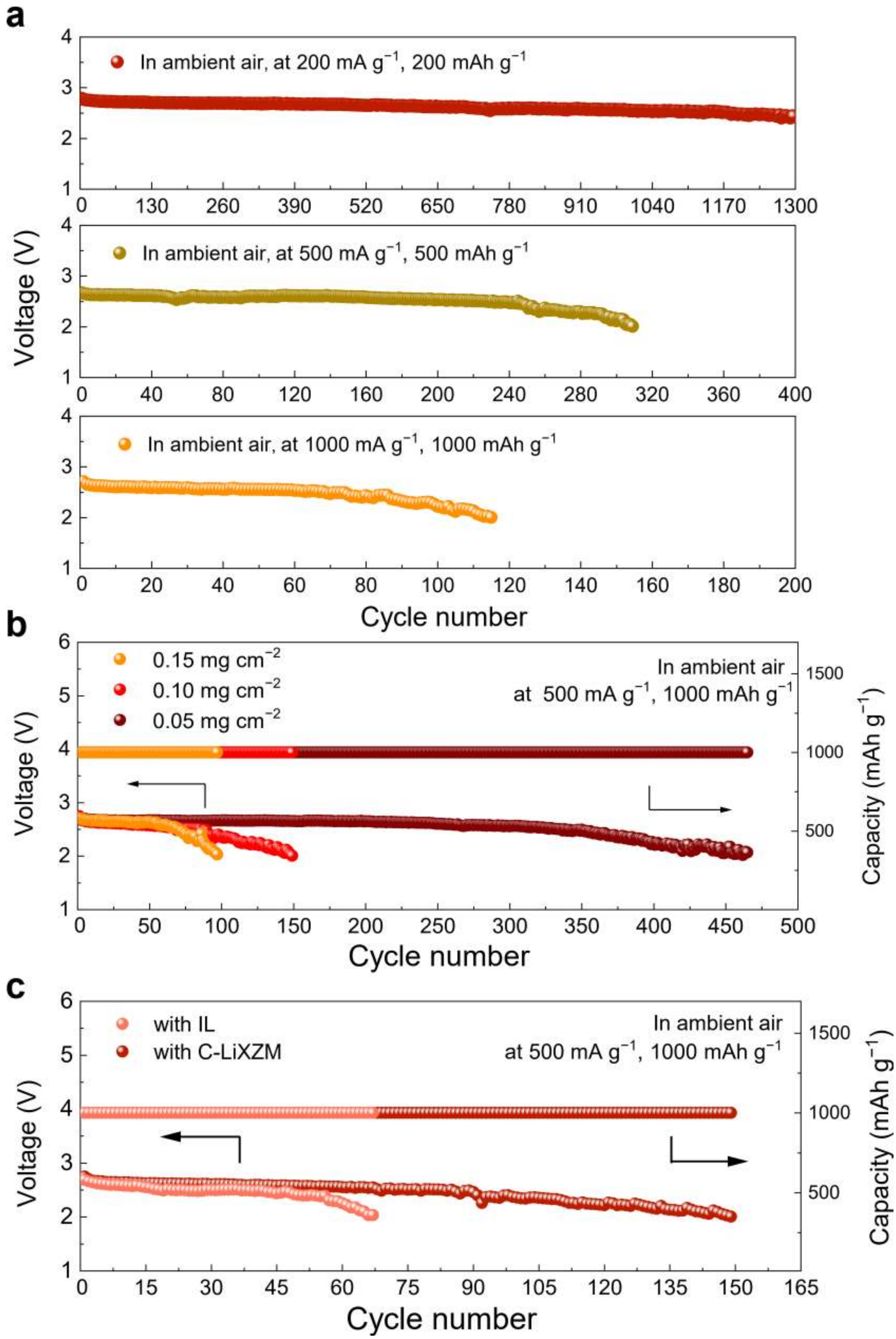
Extended Data Fig. 3 | Construction of the integrated SSLAB with C-LiXZM. **a**, Elemental mapping (C, N, O) of CNT-SS using EDS (scale bar, 200 μm). **b**, SEM image of the crystal seeds wiped on the CNT-SS (scale bar, 1 μm). **c**, Raman spectra of CNT-SS with 300 s plasma treatment before and after LiXZM growth. The two peaks located at 1,351 and 1,580 cm^{-1} arise from the D and G bands of CNT-SS. The intensity ratio of the two peaks (I_D/I_G) remained approximately the same after the in situ growth of LiXZM, indicating that the structural integrity of the CNTs was preserved. **d**, XPS survey spectrum of C-LiXZM. XPS analysis of CNT-SS demonstrates the existence of C, O, and N on the surfaces of the CNTs, whereas peaks corresponding to Si 2*p*, Al 2*p* and Li 1*s* are observed in the XPS spectrum of C-LiXZM owing to the presence of LiX zeolite. **e**, XPS spectra of C-LiXZM in the C 1*s* region. The peaks at 284.6, 285.2, 287.3 and 289.0 eV correspond to C–C (sp^3), C–N (sp^3), C=O (sp^2) and O–C=O (sp^2), respectively. **f**, XPS spectra of C-LiXZM in the N 1*s* region. The peaks of pyridinic N at 398.6 eV, pyrrolic N at 399.7 eV and graphitic N at 401.1 eV are identified in the N 1*s* spectrum of C-LiXZM. **g**, Thermogravimetric curve of C-LiXZM after the stainless-steel mesh substrate was removed. The dehydration of zeolites (<200 °C)[46] and the thermal decomposition of CNTs (500–630 °C)[47] take place in sequence as the temperature increases. **h**, Cross-sectional SEM image of the interface between LiXZM and Li metal anode (scale bar, 100 μm).



Extended Data Fig. 4 | Construction and performance of SSLABs. **a**, SEM image of the 500-mesh stainless-steel mesh as the substrate (scale bar, 50 µm). **b**, SEM image of LiXZM grown on the 500-mesh stainless-steel mesh (scale bar, 50 µm), which serves as the solid electrolyte for non-integrated batteries with C|LiXZM. **c**, XRD pattern of the commercial NASICON-type solid electrolyte of LAGP, which has a rhombohedral structure with space group[36] *R*3*c*. **d**, The first discharge–charge curves of SSLABs with C-LiXZM, C|LiXZM and C|LAGP. The curves are additionally labelled at an areal current density of 0.065 mA cm^{-2} with a limited areal specific capacity of 0.13 mAh cm^{-2}. **e**, Rate performances of the three SSLABs.

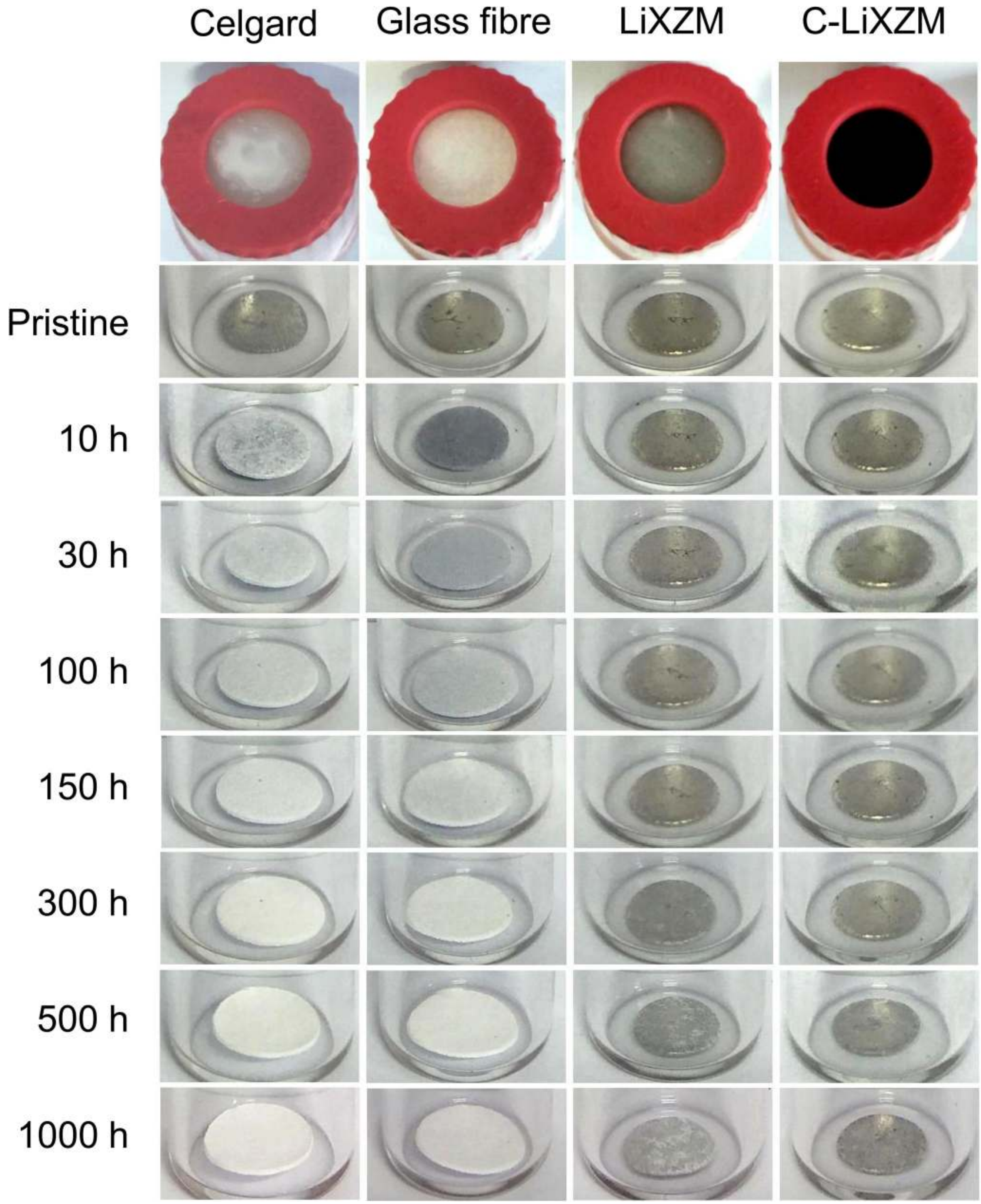


Extended Data Fig. 5 | Stability of LAGP and LiXZM. a, **b**, SEM image of LAGP before cycling (scale bar, 2 µm) (**a**) and after 10 cycles (scale bar, 2 µm) (**b**). **c**, **d**, SEM image of C-LiXZM before cycling (scale bar, 500 nm) (**c**) and after 100 cycles (scale bar, 500 nm) (**d**). The original morphology of LAGP was not maintained after cycling, whereas the outline of the crystal face of LiXZM is still clear. **e**, **f**, XRD patterns of LAGP (**e**) and LiXZM (**f**) after cycling. The crystallinity of LAGP decreased considerably, indicating the poor stability of LAGP, whereas no structural change was observed in LiXZM even after 100 cycles. **g**, **h**, Ge 3*d* XPS spectra of LAGP before (**g**) and after (**h**) cycling. **i**, **j**, Al 2*p* XPS spectra of C-LiXZM before (**i**) and after (**j**) cycling. **k**, **l**, Si 2*p* XPS spectra of C-LiXZM before (**k**) and after (**l**) cycling. When in contact with the Li metal surface, Ge^{4+} in LAGP is easily reduced, whereas no structural change occurs in LiXZM even after 100 cycles. **m**, **n**, Al 2*p* (**m**) and Si 2*p* (**n**) XPS spectra of the lithium anode before and after cycling. No Al and Si species appeared on the surface of lithium metal after cycling, which further confirmed the stability of LiXZM towards lithium metal. **o**, Results of scratch and Vickers hardness tests of LiXZM and LAGP.

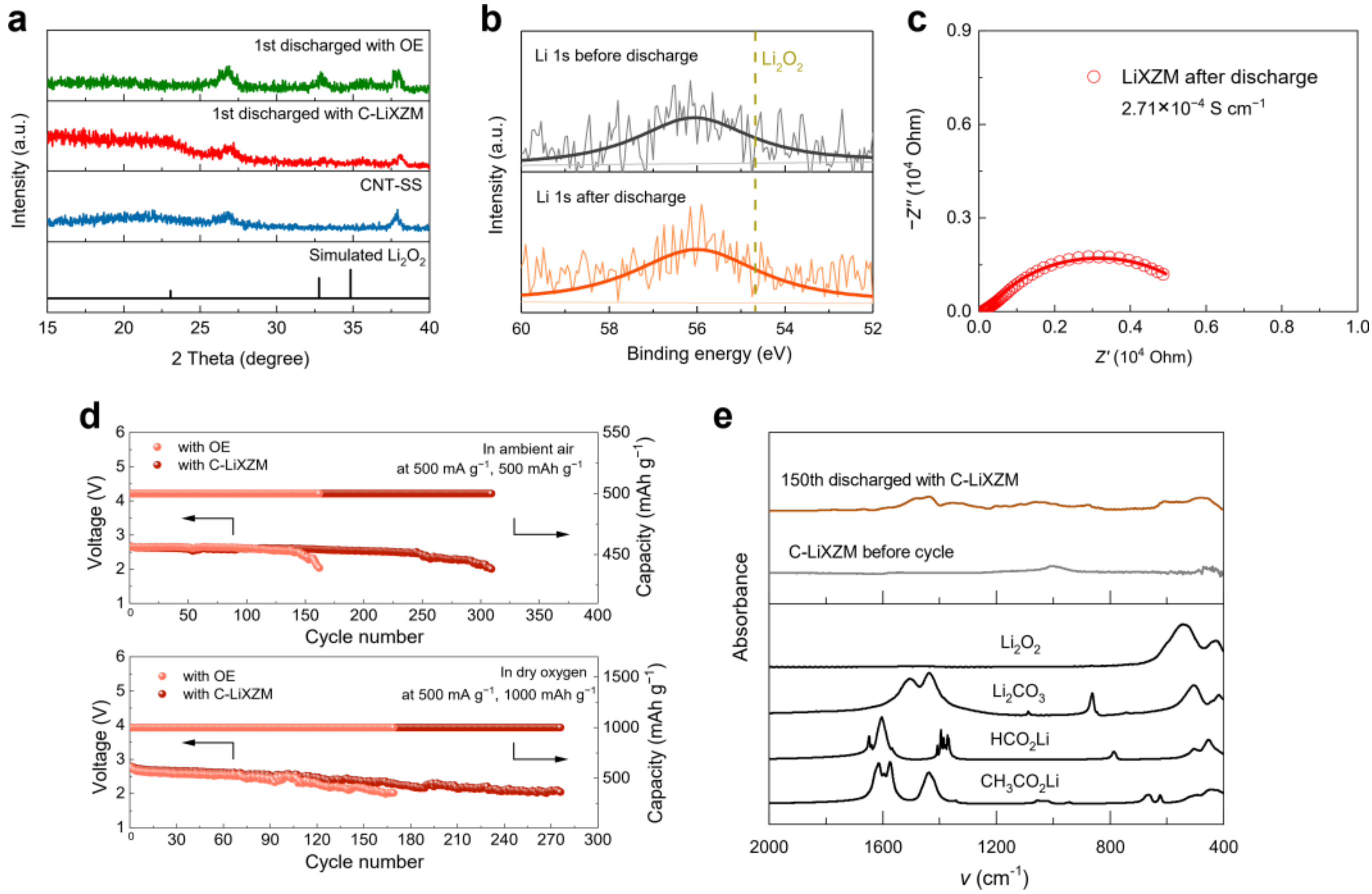


Extended Data Fig. 6 | See next page for caption.

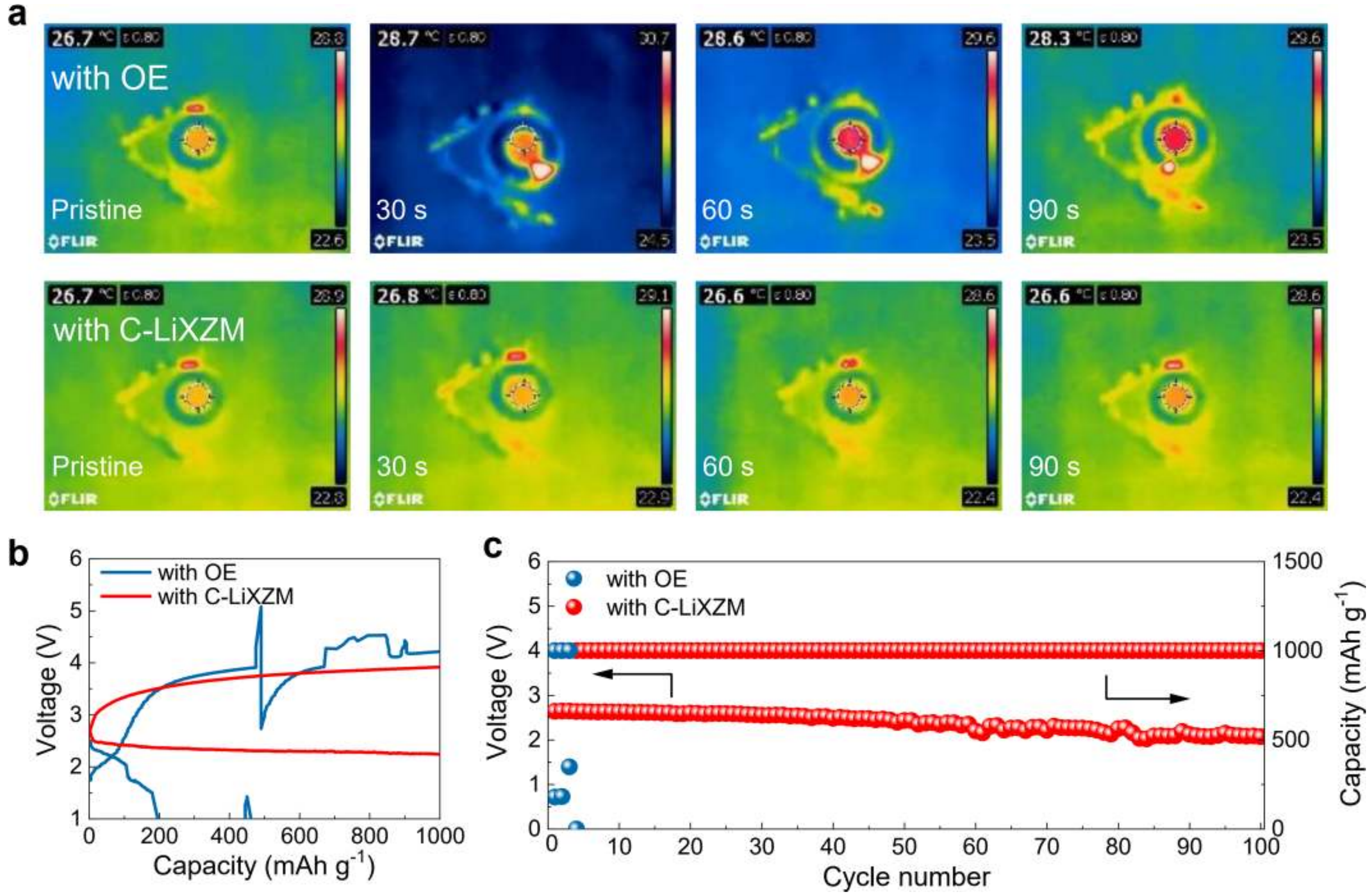
Extended Data Fig. 6 | Cycling performance of SSLABs with C-LiXZM. **a**, Cycling performance of SSLAB with C-LiXZM at 200 mA g^{-1} and 200 mAh g^{-1} (top), at 500 mA g^{-1} and 500 mAh g^{-1} (middle) and at 1,000 mA g^{-1} and 1,000 mAh g^{-1} (bottom) in ambient air. The SSLABs with C-LiXZM exhibit stable discharge–recharge reactions for 1,297 cycles at 200 mA g^{-1} and 200 mAh g^{-1}, 309 cycles at 500 mA g^{-1} and 500 mAh g^{-1}, and 115 cycles at 1,000 mA g^{-1} and 1,000 mAh g^{-1} in ambient air, and the battery keeps operating steadily at 200 mA g^{-1} and 200 mAh g^{-1} as of the submission of this work. To our knowledge, these are the best cycling performances reported to date in SSLABs. **b**, Cycling performance of SSLABs with CNT mass loadings of 0.05, 0.10 and 0.15 mg cm^{-2} at a current density of 500 mA g^{-1} with a specific capacity limited to 1,000 mAh g^{-1}. The overpotentials of the batteries display negligible change, whereas the cycle life is affected by the CNTs mass loading. Specifically, SSLABs using C-LiXZM could operate for 465, 149 and 96 cycles with CNT mass loadings of 0.05, 0.10 and 0.15 mg cm^{-2}, respectively. The reduced cycle life for the cathode materials with increased mass loading is common in practical battery systems, because the accumulation of active substance affects mass transfer[48]. In the meantime, the absolute current of the battery under the same current density would increase with increased active mass loading. **c**, Terminal discharge voltage of the SSLAB with C-LiXZM compared with the Li–air battery with ionic liquid electrolyte (IL). The cyclic stability of the SSLAB with C-LiXZM is better than that of the Li–air battery with ionic liquid electrolyte, which operates for only 67 cycles before failure—mainly as a result of similar problems to the organic-electrolyte-based Li–air batteries, including the decomposition of the ionic liquid and growth of Li dendrites[49].



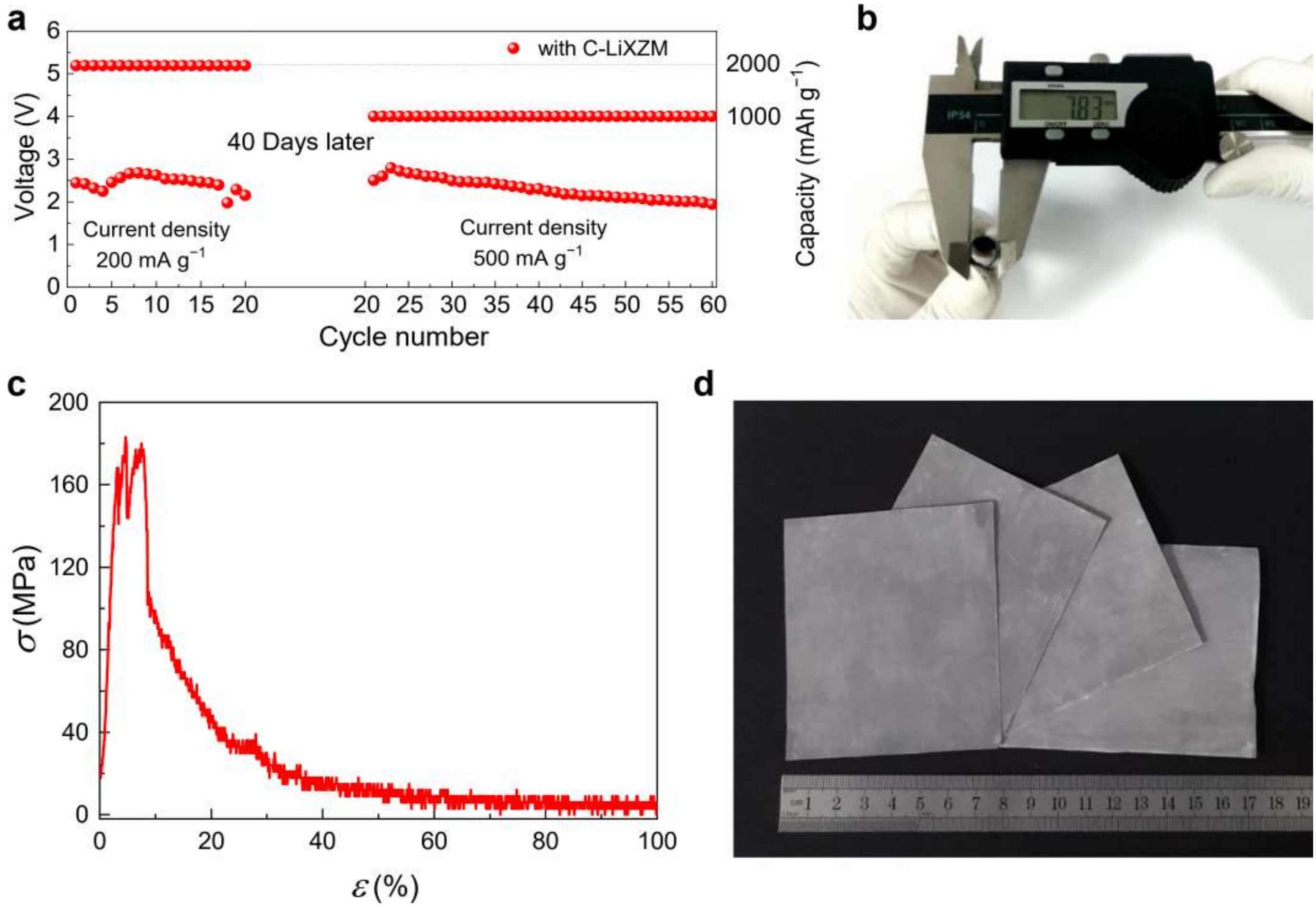
Extended Data Fig. 7 | Protective effect of C-LiXZM. Time-resolved optical images of lithium metal inserted in the test devices sealed with Celgard separator (immersed with organic electrolyte), glass fibre separator (immersed with organic electrolyte), LiXZM and C-LiXZM. The lithium metal sealed with LiXZM and C-LiXZM retains its metallic lustre after 1,000 h of ageing, which demonstrates that lithium metal can be effectively protected from the erosion of air components by zeolite membranes. Notably, lithium metal sealed with C-LiXZM is brighter than that sealed with LiXZM, because the hydrophobic carbon cathode surface can further block H_2O from the air.



Extended Data Fig. 8 | Characterization and properties of LiXZM. a, XRD patterns of the discharge products of batteries with C-LiXZM and the organic electrolyte after the first discharge. **b**, Li 1*s* XPS spectra of LiXZM before and after cycling. The surface of the LiXZM adjacent to the cathode side was characterized by XPS after cycling, and no discharge products of Li_2O_2 were detected. **c**, EIS spectrum of LiXZM after discharge. The ionic conductivity after discharge is 2.71×10^{-4} S cm^{-1}, showing no notable difference from that before discharge (2.65×10^{-4} S cm^{-1}). **d**, Terminal discharge voltage of the SSLABs with C-LiXZM compared with those of the Li–air batteries with organic electrolyte at 500 mA g^{-1}, 500 mAh g^{-1} in ambient air (309 versus 162 cycles), and 500 mA g^{-1}, 1,000 mAh g^{-1} in dry oxygen (276 versus 169 cycles). **e**, FTIR spectra of the discharge products in the SSLAB with C-LiXZM after 150 cycles.



Extended Data Fig. 9 | The nail test on the batteries with C-LiXZM and organic electrolyte. a, The short-circuit battery temperature observed through an infrared camera. **b**, The first discharge–charge curves of the batteries with organic electrolyte and C-LiXZM after the nail test. **c**, Terminal discharge voltage of the batteries with organic electrolyte and C-LiXZM after the nail test at a current density of 500 mA g^{-1} with the specific capacity limited to 1,000 mAg^{-1}. The nail pierced the batteries containing C-LiXZM and organic electrolyte powering a red LED (Supplementary Video 1, 2). The battery with C-LiXZM was able to light up the LED immediately after the nail was removed, whereas the battery with organic electrolyte was no longer able to function. The temperature of the battery with organic electrolyte immediately increased by 2 °C after the short circuit, whereas no temperature change was observed in the case of SSLABs with C-LiXZM. It is notable that a very small button battery with an open system produced such a temperature difference. If a battery pack that contains flammable and explosive organic electrolytes is hit or damaged when used in a vehicle, a short circuit could be very dangerous.



Extended Data Fig. 10 | Potential for the practical application of zeolite membranes as solid electrolytes. a, The performance of the SSLAB with C-LiXZM under a simulated actual application state. The integrated SSLAB with C-LiXZM was operated for 20 cycles (200 mA g^{-1}, 2,000 mAh g^{-1}). After an interval of 40 days, the battery with C-LiXZM was able to operate steadily for 40 cycles at a different current density and specific capacity (500 mA g^{-1}, 1,000 mAh g^{-1}). **b**, Curvature test of the integrated SSLAB with C-LiXZM. The curvature radius and curvature are 3.92 mm and 2.55×10^2 m^{-1}, respectively. **c**, Stress strain curve (σ–ε) of the C-LiXZM. **d**, A digital photograph of large LiXZMs synthesized in the laboratory.